GURDHARAM JEET SINGH

Software Engineer Intern • Full-Stack & Web Systems

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EDUCATION

Baba Farid Group of Institutions (BFGI) **Expected May 2028**

Bachelor of Science (Hons.) • Applied Computational Systems Focus *Punjab, India*

Coursework: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Systems (DBMS), Computer Networks

Senior Secondary Education (Class XII / Grade 12) — 91.0% (Academic Distinction) **Completed 2024**

Physics, Chemistry, Mathematics (PCM) • Quantitative Foundations *Punjab, India*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL (PostgreSQL), HTML5, CSS3, Git, Bash

Frontend: React, Next.js, Tailwind CSS, Vite, Three.js, WebGL, Canvas API, Framer Motion, Responsive Design

Backend & Systems: Node.js, Express, PostgreSQL, REST APIs, WebSockets, Docker, OOP, Testing (Pytest, Jest)

AI & Cloud Tools: LLMs, AI Agents, Gemini API, Meta Cloud API (WABA), GitHub Actions CI/CD, Railway

PROJECTS

EduCore — College Management ERP & Student Information System **2025 – Present**

React, TypeScript, Express, PostgreSQL, MariaDB Connection Pools, Railway github.com/gurination1/ERP_educore

- **Campus Administration & Academic Management:** Built full-stack college ERP managing student admissions, academic grading, and fee reconciliation across campus departments.
- **Database Abstraction & RBAC:** Implemented pooled PostgreSQL and MariaDB database abstraction with automated connection health checks and 9-role JWT access controls.

yt-auto — 24/7 Autonomous Cloud Media Production Fleet **2025 – Present**

Python, FFmpeg, GitHub Actions CI/CD Matrix, Gemini Multimodal AI github.com/gurination1/yt-auto

- **Distributed Cloud Processing:** Built automated media pipeline orchestrating batch video generation across parallel GitHub Actions matrix runners with zero dedicated server costs.
- **Automated QA & Audio Mastering:** Automated video rendering with custom FFmpeg filtergraphs and sidechain compression, integrating a Gemini AI evaluation check to filter visual artifacts.

Fasal Doctor — Autonomous Edge-AI Foliar Pathology Platform **2026 – Present**

Flutter, Dart, TensorFlow Lite, INT8 Post-Training Quantization, Worker Isolates Mobile System

- **On-Device Offline Neural Inference:** Deployed INT8 quantized MobileNet foliar pathology models running 100% offline on low-power mobile devices with sub-50ms inference.
- **Asynchronous Thread Architecture:** Implemented Flutter background worker isolates to decouple camera frame previews (60 FPS) from tensor inference, preventing UI freezes.

Spatial & Creative Web Suite — SOLUM Minerals, Dream Heights & Branders **2025 – Present**

Next.js, Three.js, WebGL, Tailwind CSS, Framer Motion, Lenis Smooth Scroll Open Source

- **Procedural In-Memory Asset Generation:** Built procedural Three.js asset pipeline synthesizing dynamic noise textures in-memory via Canvas API (SOLUM Minerals), reducing static asset bundle size by 85%.
- **Hardware Acceleration & Web Vitals:** Developed agency portals (Branders) and spatial views (Dream Heights) with Lenis momentum scroll, maintaining stable 60 FPS with zero layout shift.

WORK EXPERIENCE

CP Systems (Cool & Pure System) **Jan 2024 – Jan 2025**

Software Developer (Contract / Part-Time) • Industrial RO Plants & Dairy Equipment (cpsystem.in) *Punjab, India*

- **Industrial Telemetry & Database Optimization:** Built operational telemetry and inventory microservices using Node.js and SQL, optimizing query plans to reduce p95 latency by 42%.
- **Meta Cloud API & CI/CD Pipelines:** Automated customer notifications over Meta Cloud API and implemented GitHub Actions CI/CD with caching, reducing build times from 14m to 3.8m.

Gurdharam Enterprise **Jan 2025 – Present**

Freelance Full-Stack Developer • Web Platforms & Client Engineering *Remote*

- **Commercial Client Delivery:** Developed production web applications, agency portals (Branders), and responsive frontend interfaces, delivering sub-second page loads.
- **Conversational Voice Pipeline:** Built low-latency voice pipeline integrating Exotel WebSockets, Silero VAD voice interruption, and streaming Sarvam AI Indic LLMs.